Prep Test 3: Functions

75 minutes | 100 points | 12 questions

Name: _____ Date: _____

Show your work. Clearly label axes and requested points. Use interval notation where requested.

A four-function calculator and one single-sided notesheet (no example problems) are allowed. No graphing calculator.

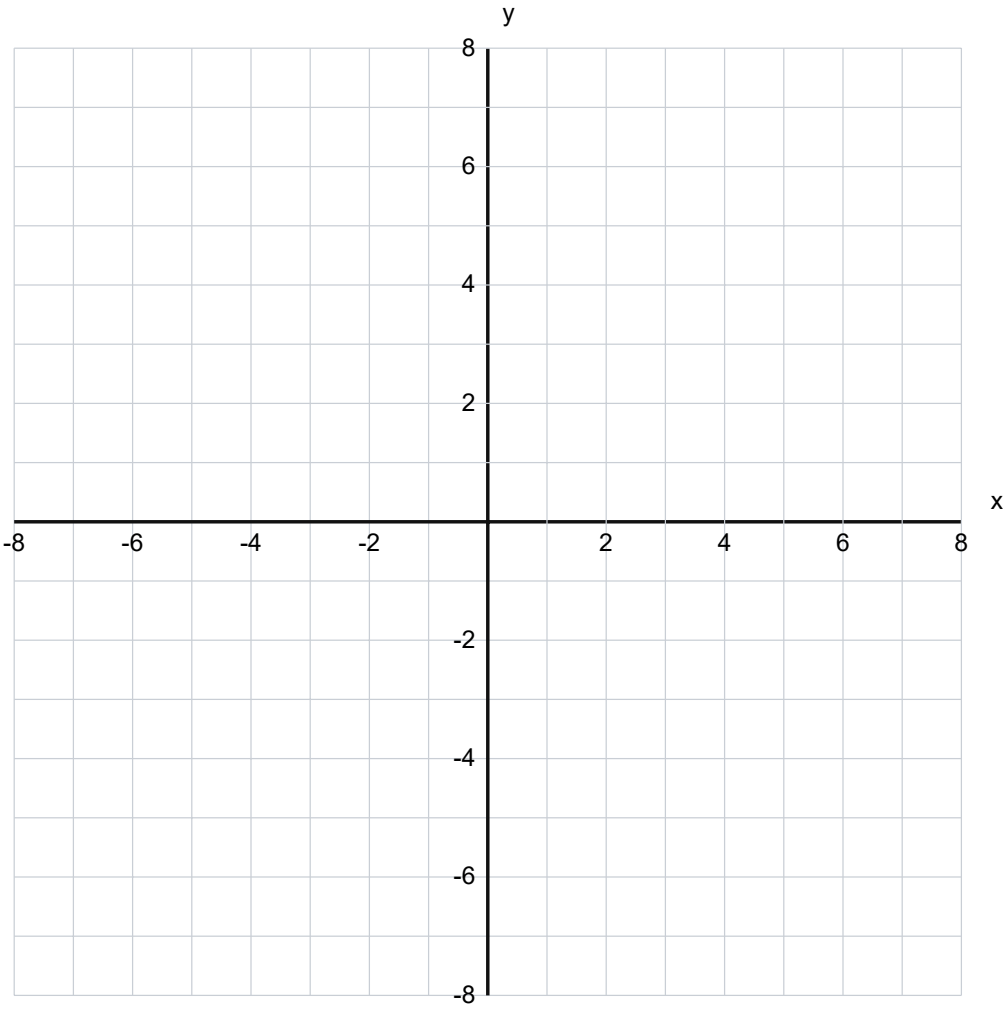
Use the blank space provided for answers and calculations. Graphs use one unit per square.

Question 1

10 points

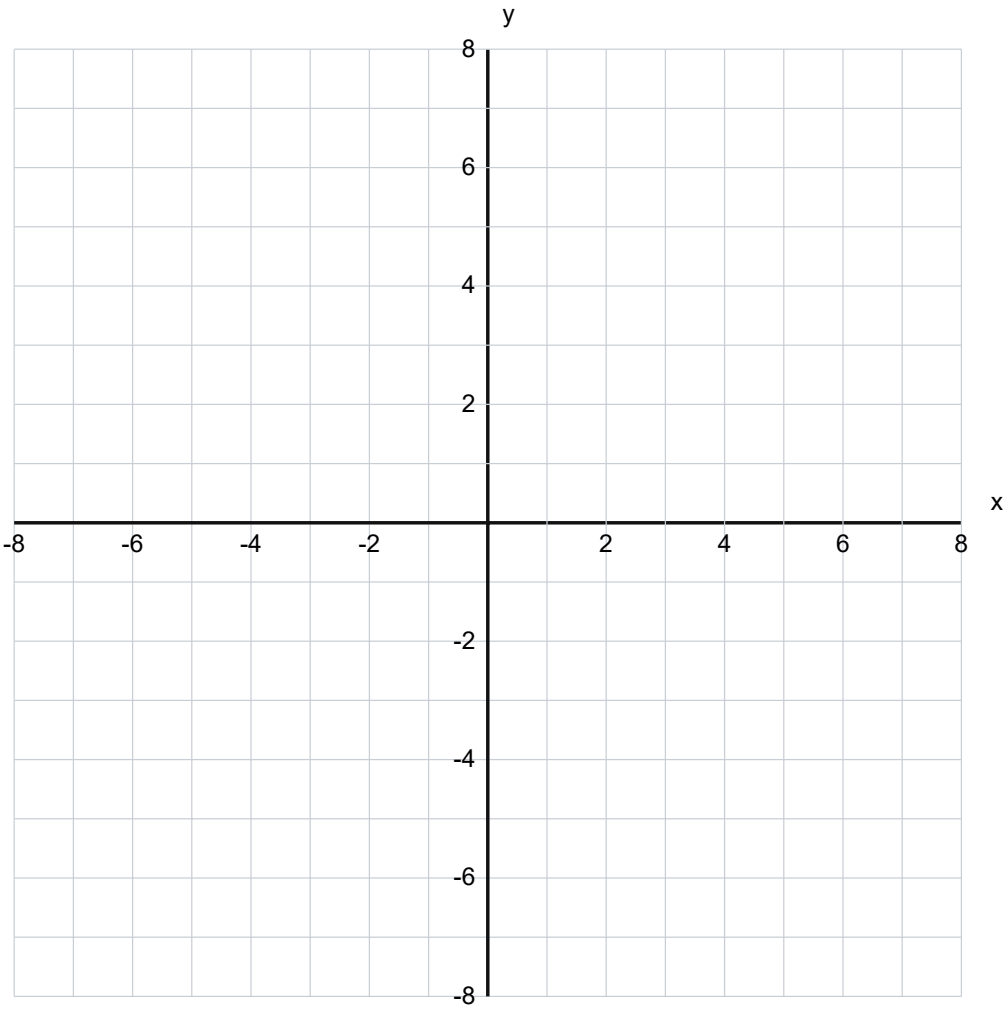
Graph each function. Label the axes and intercepts.

(a) $f(x) = \frac{3}{2}x - 2$.



Question 1, continued

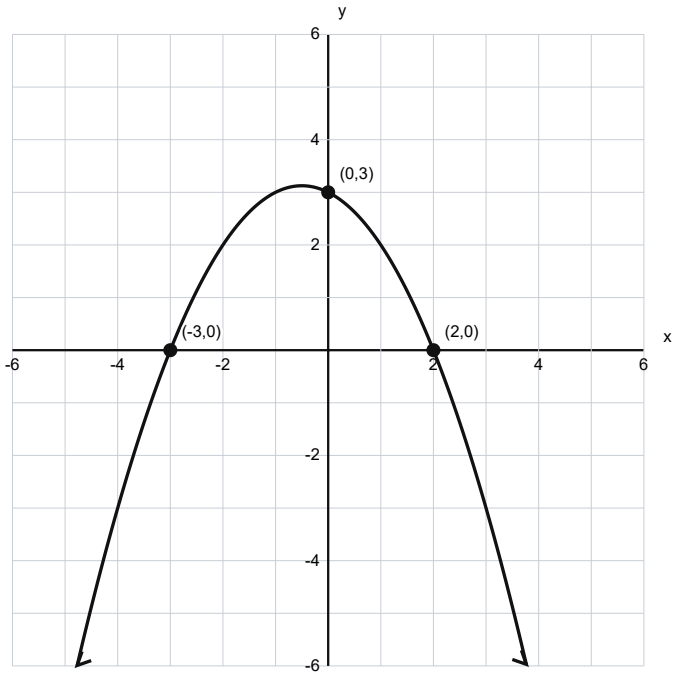
(b) $g(x) = -\frac{3}{2}\sqrt{-x+2}+3$.



Question 2

6 points

Write an equation for the graphed function in factored and standard form. Show your work.

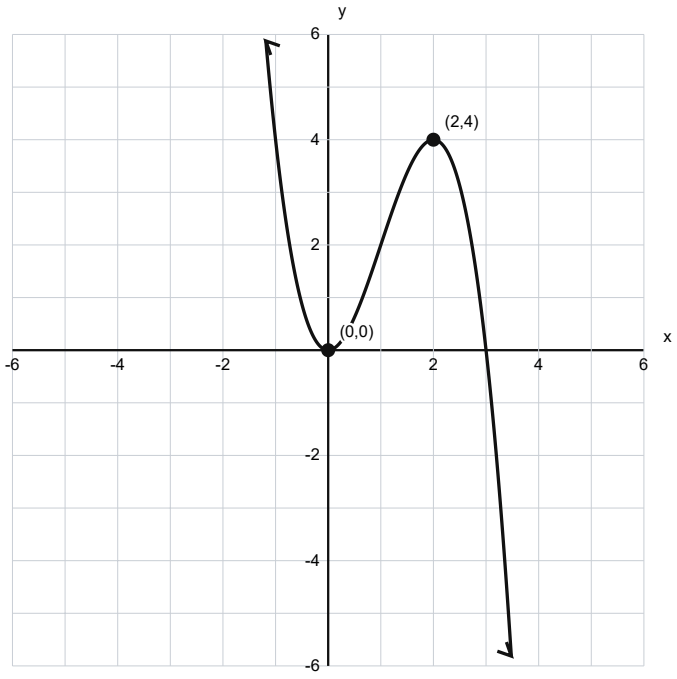


Given graph

Question 3

10 points

For the graphed function, give the domain, range, end behavior, intervals of increase and decrease, and relative extrema. Use proper notation.



Given graph

Domain:

Range:

End behavior:

Increasing intervals:

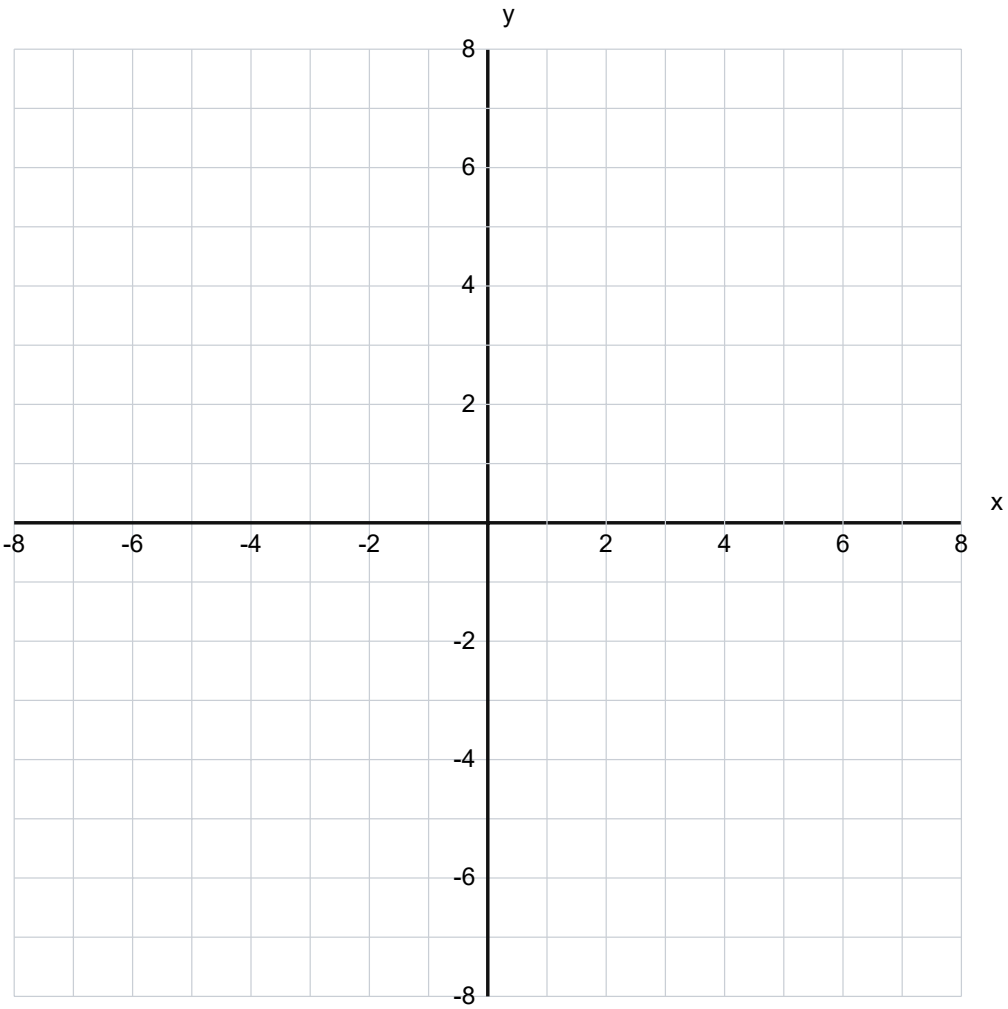
Decreasing intervals:

Relative extrema:

Question 4

6 points

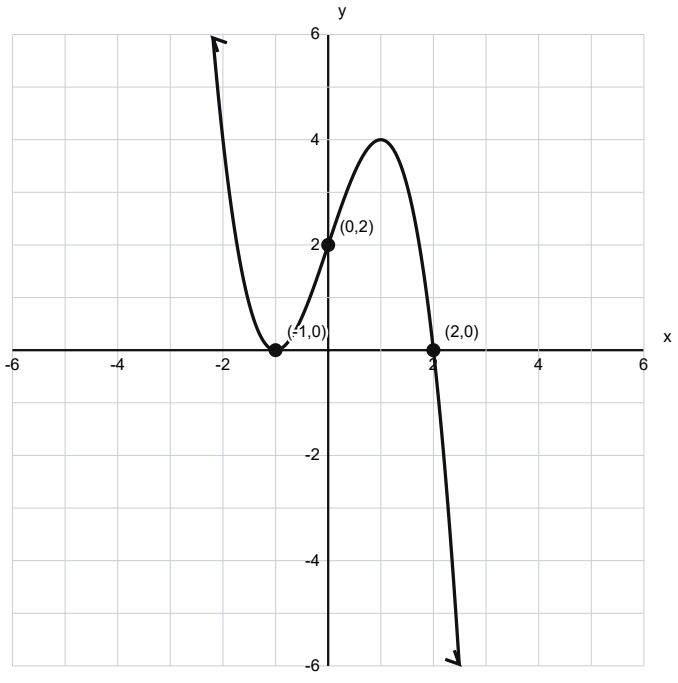
Graph $F(x) = -x^3 + x^2 + 6x$. Label the axes and intercepts.



Question 5

6 points

Write an equation for the graphed polynomial. Show your work.



Given graph

Question 6

12 points

Let $f(x) = \sqrt{8 - 2x}$ and $g(x) = x + 1$.

(a) Find $\frac{f(x)}{g(x)}$ and its domain.

(b) Find $g(f(x))$.

(c) Find $f(a - b)$.

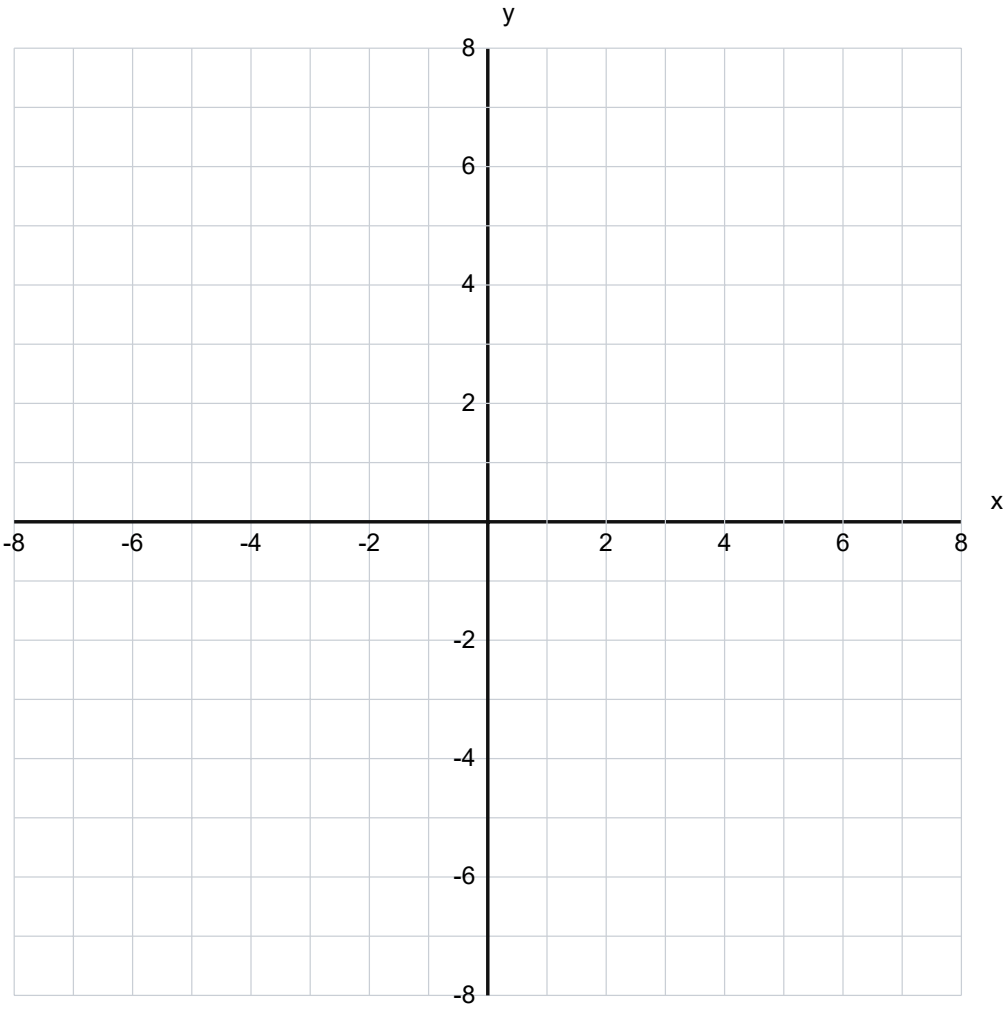
Question 7

12 points

Let $h(x) = \begin{cases} 2(x+1)^2 - 4, & x < -1 \\ -2x + 1, & x \geq -1. \end{cases}$

(a) Graph h on the provided axes.

(b) Find $h(-2)$, $h(-1)$, and $h(2)$.



Question 8

12 points

A print studio makes custom fabric banners. A banner with an area of up to 6 square feet costs a flat $24. For larger banners, the studio charges $24 plus $3.50 for each square foot beyond the first 6; fractional square feet are charged proportionally. The studio makes banners with areas greater than 0 and no larger than 30 square feet.

(a) Write a piecewise function C(a) for the cost in dollars, where a is the banner area in square feet.

(b) Find the costs for banners with areas of 6, 10, and 30 square feet.

Question 9

8 points

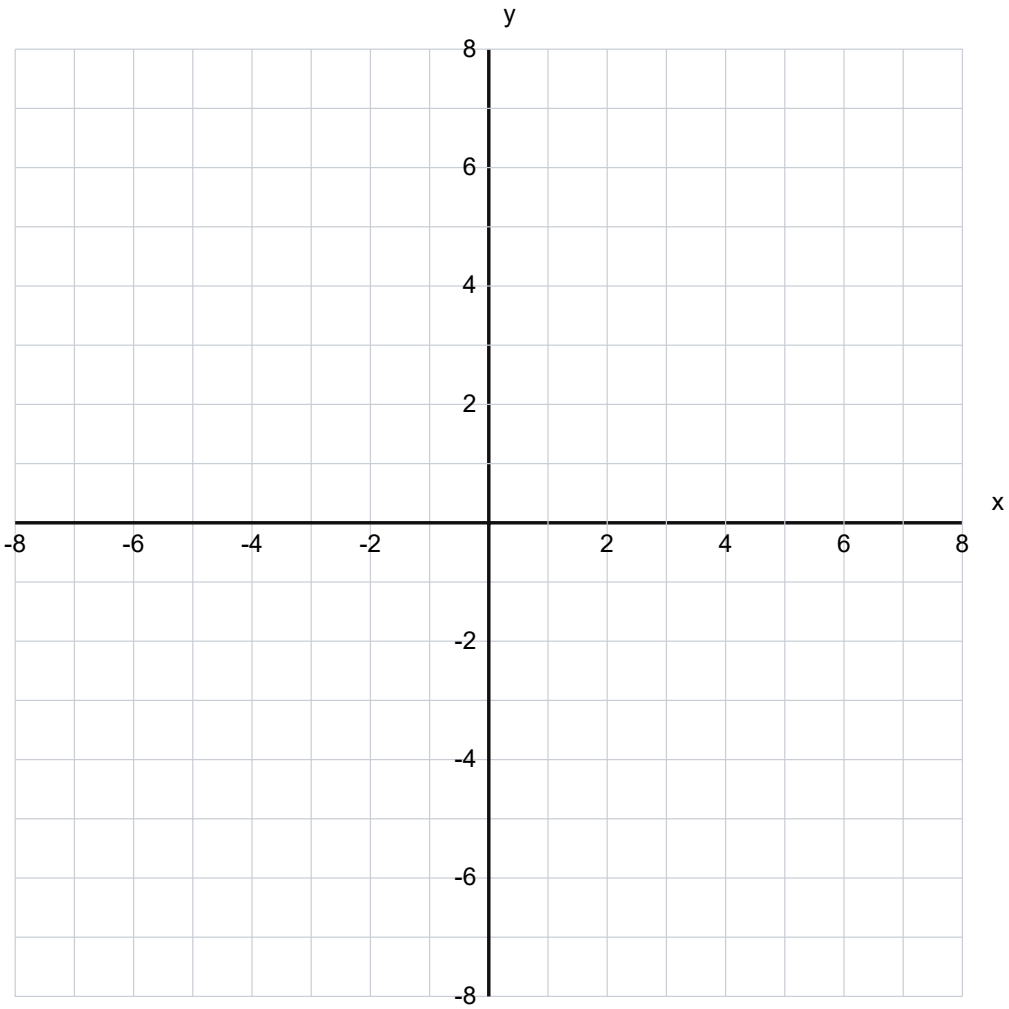
Graph a function with all of the following properties:

As $x \to -\infty$, $y \to \infty$, and as $x \to \infty$, $y \to \infty$.

Decreasing on $(-\infty, -2)$.

A relative minimum at $(2, -2)$.

Y-intercept $(0, 2)$.



Question 10

8 points

Let $j(x) = \frac{5}{x^2+4}$. Find two functions f and g such that $j(x) = f(g(x))$. Do not use $f(x) = x$ or $g(x) = x$.

Show that your functions give j when composed.

Question 11

4 points

The continuous function g is defined for all real numbers. The table gives some values and its behavior on the indicated intervals.

Which statement is guaranteed to be true?

(A) g has a relative maximum at x=4.

(B) The range of g is [-2,8].

(C) g(x)=0 for some x in (4,7).

(D) g has a relative minimum at x=0.

Behavior of g

x	**-5**	**(-5,0)**	**0**	**(0,4)**	**4**	**(4,7)**	**7**
g(x)	6	decreasing	-2	increasing	3	increasing	8

Question 12

6 points

The graph of g is shown for $-4 \leq x \leq 8$. Selected values of f are in the table; f is defined for all real numbers. Let $h(x) = 3x + 2$.

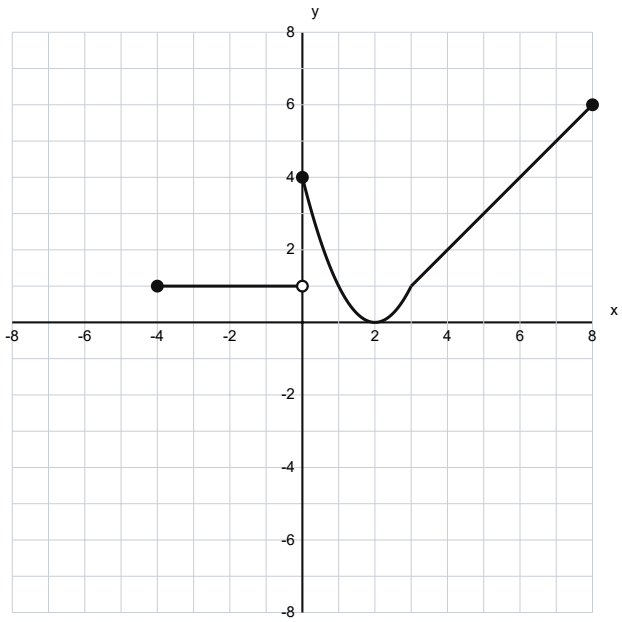
(a) Find $f(g(-2))$.

(b) Let $j(x) = g(x) - h(x)$. Find $j(5)$.

(c) What is the domain of $\frac{f(x)}{h(x)}$?

Selected values of f

x	-1	1	2	4	6
f(x)	5	-3	7	2	0



Given graph